

54045. Technologies

BA SKA B5 / BA KUG B5 / BA InAS D5/D6

WINTER 2026/27

Mondays, 14-16h

Venue: S6 (GWII)

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History

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Description

The seminar is an introduction to the field of Science and Technology Studies (STS). STS is dedicated to the social science analysis of the interplay between knowledge, technology and society. Technologies have not only made industrial society possible and shaped modernity, but are also closely linked to its excesses (e.g. environmental pollution, climate change, post-truth politics). Technologies also dominate and enrich our daily lives and social interactions.

This semester (Winter 2026/7), the course is co-taught by an anthropologist of technology and a cultural historian of sound. Across the semester, we will trace the dynamics between technology, society, and sound as much through the history and performance of sound as through computers and laboratories. We will ask how algorithms compute and shape musical taste; how instruments, recordings, and performance — from Bach's solo violin works to the Edison phonograph's pursuit of sonic "fidelity" — mediate what counts as musical authenticity. We explore the overlooked work of women in electronic music while examining science and technology themselves as staged and performative, from Latour's laboratory theater to the Margravia Opera House here in Bayreuth. Finally, we consider how mechanical (re)production — in manuscript, print, photography, and code — has reshaped aesthetic judgment, authorship, and authenticity, from medieval scribes and early modern printing to Walter Benjamin's reflections on art in the age of mechanical reproduction.

Based on empirical case studies we will examine different theoretical and methodological approaches to the relationship between knowledge (science), technology and society. We will address the following questions, among others:

- How does science create knowledge? How can a social science analysis of knowledge (science) practices look like?
- How are design, dissemination, application, and appropriation of technologies interconnected?
- What kinds of societies and politics do certain scientific knowledge, technology and technical infrastructures enable?

- What specific questions and problems arise for an STS that focuses on the global South?

This seminar will be taught in English. Some texts are available in both German and English.

Exam (Hausarbeit)

As a textual form, manifestos have a long history of intervening in debates about technology and society — from technical communities issuing statements of principle, to critical and artistic responses like Donna Haraway's "A Cyborg Manifesto," which embraces the politics of technoscientific more-than-humans, to F.T. Marinetti's "Manifesto of Futurism," which calls for the embrace of the modern machine age. A manifesto declares an ethical, political, or social position. We encourage you to experiment with this genre in the final exam.

The final exam is a long-form essay (Hausarbeit) containing two sections. Section 1 requires you to write a manifesto. Section 2 (requires you to explain the manifesto you wrote by reflecting on the themes of the course.

Important Information

Following Para. 11, no. 8 of the [Studien- und Prüfungsordnung](#) (p. 10):

- Section 1 should be maximum 3 pages.
- Section 2 can be between 3-15 pages.
- Your manifesto can be written in German or English. You are encouraged to write your reflection in English, and you are allowed to use large language models (LLM) for translation (with acknowledgement).
- Due 30 March with feedback. Due 30 April without feedback.

In your long-form essay, please answer the following questions:

Section 1 (max. 3 pages): Write a manifesto on the student use of A.I. at the University of Bayreuth. Please take a clear position.

You may take any position:

- ***In favor*** — arguing for AI's expansion, adoption, or a particular use of it
- ***Against*** — arguing for resistance, restriction, or rejection
- ***Neither*** — refusing the binary, reframing the question, or making a different kind of argument (e.g., about who should control AI, or on what terms, rather than whether it should be used)

Section 2 (3-15 pages): Reflect on the manifesto you just wrote, linking your reflections to readings and discussions from the course. Do not summarize — it is

an analysis of *why* you wrote the manifesto the way you did and *how* it relates to what we've read and discussed this semester.

Section 2 should address some of these points, which are meant to help guide you in your writing:

- **Position and its sources.** Where does your stance on AI come from? Which course readings, concepts, or class discussions influenced, inspired or informed your position, including the ones you disagreed with?
- **Genre and rhetoric.** What choices did you make in writing the manifesto (tone, structure, imagery, audience) and why? Who is the target audience of your manifesto? Anthropology and history students at UBT? The entire campus? All of Germany's or Europe's youth? What did the manifesto form let you say, or force you to leave out, that a more conventional essay would not?
- **Conceptual framing.** Situate your own manifesto within the theoretical vocabulary of the course — for example, does it treat technology as autonomous and deterministic, or as socially shaped and contested? Does it invoke specific actors, infrastructures, or histories we've discussed? Would you categorize your own manifesto as an example of a pattern we've studied (automation, authenticity, equality, nation-building, humanity)?
- **A moment of friction.** Do you think your manifesto is persuasive? Why or why not? Identify at least one course reading or discussion that complicates or unsettles your manifesto's position, and engage with it seriously rather than dismissing it.

Essays and the cumulative mark for the module will be graded according to §16 of the [Prüfungs- und Studienordnung für den Bachelorstudiengang Sozial- und Kulturanthropologie an der Universität Bayreuth](#) (link is to 2021 version, but please consult the latest version where available).

Accessibility and Inclusion

We are committed to partnering with you on your academic and intellectual journey. We recognize that your ability to thrive academically can be impacted by your personal well-being and that stressors may impact you over the course of the semester. If the stressors are academic, we welcome the opportunity to discuss and address those stressors with you in order to find solutions together. If you are experiencing challenges or questions related to emotional health, finances, physical health, relationships, learning strategies or differences, or other potential stressors, we encourage you to reach out to [resources available at the University](#).

If, at any point in the semester, a disability or personal circumstances affect your learning in this course or if there are ways in which the overall structure of the course and general classroom interactions could be adapted to facilitate full participation, please do not hesitate to reach out to me. It is never too late to contact us about this – our bodies and

circumstances are continuously changing. However, I encourage you to contact us in advance of deadlines so that we can plan together.

Helpful Resources (available at ZB)

- Sismondo, Sergio. *An Introduction to Science and Technology Studies*. 2nd ed. Chichester, West Sussex, U.K. ; Malden, MA: Wiley-Blackwell, 2010.
- Bauer, Susanne, Torsten H. Voigt, and Thomas Lemke, eds. *Science and technology studies: klassische Positionen und aktuelle Perspektiven*. 2. Auflage. Suhrkamp-Taschenbuch Wissenschaft 2193. Berlin: Suhrkamp, 2020.
- Felt, Ulrike, ed. *The Handbook of Science and Technology Studies*. Fourth edition. Cambridge, Massachusetts: The MIT Press, 2017.
- Beck, Stefan, Jörg Niewöhner, and Estrid Sørensen. *Science and Technology Studies: Eine Sozialanthropologische Einführung. Verkörperungen/MatteRealities*, Band 17. Bielefeld: Transcript, 2012.
- Sterne, Jonathan, ed. 2012. *The Sound Studies Reader*. Routledge.

SEMINARPLAN

subject to change

all changes will be announced in class and on e-learning

Week 1 (19. Oct): What is Technology?

Introduction to the course and syllabus.

Reading:

- None

Week 2 (26. Oct): The Moveable Type Press

Reading:

- Beach, Alison I., Erik Kwakkel, Peter Vandenabeele, Eva Vermeersch, and Sophie Andrade. 2026. "Guda, a Sinful Woman: A Multi-Scalar Portrait of a Medieval Scribe and Illuminator." *Historical Research* 99 (284): 204–34.
<https://doi.org/10.1093/hisres/htaf017>.
- Palabıyık, Nil. 2015. "The Beginnings of Printing in the Ottoman Capital: Book Production and Circulation in Early Modern Constantinople." *Studies in Ottoman Science* 16 (2): 3–32. <https://dergipark.org.tr/en/download/article-file/673531>.

Further Reading:

- Marx, Leo. 1997. "‘Technology’: The Emergence of a Hazardous Concept." *Social Research* 64 (3): 965–88.
- Pfaffenberger, Bryan. 1992. "Social Anthropology of Technology." *Annual Review of Anthropology* 21: 491–516.

Week 3 (2. Nov): Computing Music Taste

Reading:

- Seaver, Nick. 2022. "Introduction: Technology with Humanity," in *Computing Taste: Algorithms and the Makers of Music Recommendation*. University of Chicago Press, pp. 1-21.
- Rasmus Fleischer, "How Music Takes Place: Excerpts from 'The Post-Digital Manifesto,'" in *e-flux journal: The Internet Does Not Exist*, ed. Julieta Aranda, Brian Kuan Wood, and Anton Vidokle (Berlin: Sternberg Press, 2015), 132–49.

Further Reading:

- Shapin, Steven. 2016. "A Taste of Science: Making the Subjective Objective in the California Wine World." *Social Studies of Science* 46 (3): 436–60.
<https://doi.org/10.1177/0306312716651346>.
- Daston, Lorraine, and Peter Galison. *Objectivity*. Cambridge, MA: Zone Books, 2007.

Week 4 (9. Nov): Technologies of Music

Reading:

- [Selections] Attali, Jacques. 1985. *Noise: The Political Economy of Music*. Theory and History of Literature, v. 16. University of Minnesota Press.
- [Selections] Bijlsma, Anner. 2012. *Bach Senza Basso: About the Solo Works for Violin of Joh. Seb. Bach*. Bylsma's Fencing Mail.

Further Reading:

- Thompson, Emily Ann. 2002. *The Soundscape of Modernity: Architectural Acoustics and the Culture of Listening in America, 1900-1933*. MIT Press.
- Helmreich, Stefan. 2013. "Seashell Sound." *Cabinet*, no. 48 (Winter 2012/13): 23–29. <https://www.cabinetmagazine.org/issues/48/helmreich.php>.

Week 5 (16. Nov): Staging Science

Reading:

- Latour, Bruno. "Give Me a Laboratory and I Will Raise the World." In *Science Observed: Perspectives on the Social Study of Science*, edited by Karin D. Knorr-Cetina and Michael Mulkay, 141–170. London: Sage Publications, 1983.
 - o Latour, Bruno. 2006. „Gebt mir ein Laboratorium und ich werde die Welt aus den Angeln heben.“ In Bellinger, Andreas und David J. Krieger (Hrsg.) *ANThology. Ein einführendes Handbuch zur Akteur-Netzwerk-Theorie*. Bielefeld: transcript: 103-34.
- Kotnik, Lia Lola V. 2016. "Opera as Social Showcase: Rituals of 'Magic Mirrors' at the Margravian Opera House in Mid-Eighteenth-Century Bayreuth." *Etnoantropološki Problemi / Issues in Ethnology and Anthropology* 11 (March): 19–45.
<https://doi.org/10.21301/EAPV11I1.1>.

Further Reading:

- Amelang, Katrin 2012. „Laborstudien.“ In *Science and Technology Studies. Eine sozialanthropologische Einführung*, hg. von Stephan Beck, Jörg Niewöhner und Estrid Sørensen, 145-71. Bielefeld: transcript.

Week 6 (23. Nov): Mechanical Production

Reading:

- William Henry Fox Talbot, *The Pencil of Nature* (London: Longman, Brown, Green, and Longmans, 1844–46; Project Gutenberg, 2010), <https://www.gutenberg.org/ebooks/33447>, accessed September 9, 2026.
- M'charek, Amade. 2020. "Tentacular Faces: Race and the Return of the Phenotype in Forensic Identification." *American Anthropologist* 122 (2): 369–80. <https://doi.org/10.1111/aman.13385>.

Further Reading:

- Sontag, Susan. 2020. *On Photography*. RosettaBooks.
- Paredes, Alyssa. 2018. "The Problem of Mechanization: Craft, Machines, and 'Centering' in a Japanese *Mingei* Pottery Village." *Journal of Material Culture* 23 (2): 133–50. <https://doi.org/10.1177/1359183517725366>.

Week 7 (30. Nov): Does Technology Have Politics?

Reading:

- Winner, Langdon. "Do Artifacts Have Politics?" *Daedalus* 109, no. 1 (1980): 121–36.
- (Watch) Franz Steiner Verlag. 2022. *Frank Uekötter Im Interview Zu Seinem Buch „Atomare Demokratie“*. 6:52. https://www.youtube.com/watch?v=3o_bu7gtOdk.

Further Reading:

- Moran-Thomas, Amy. "How a Popular Medical Device Encodes Racial Bias." *Boston Review*, August 2, 2020. <http://bostonreview.net/science-nature-race/amy-moran-thomas-how-popular-medical-device-encodes-racial-bias>.

Week 8 (7. Dec): Technology, Users, and Design

Reading:

- Latour, Bruno. 2000. "The Berlin key or how to do words with things." In Graves-Brown, Paul, ed. *Matter, Materiality, and Modern Culture*. 1. publ. London: Routledge, 2000.
 - o Latour, Bruno. 1996. *Der Berliner Schlüssel: Erkundungen eines Liebhabers der Wissenschaften*. Akad.-Verl.
- Emily Thompson, "Machines, Music, and the Quest for Fidelity: Marketing the Edison Phonograph in America, 1877–1925," *The Musical Quarterly* 79, no. 1 (Spring 1995): 131–71.

Further Reading:

- Akrich, Madeleine. 1992. "The De-Description of Technical Objects" In *Shaping Technology/Building Society*, ed. by Wiebe Bijker and John Law, 205–24. Cambridge, Mass.: MIT Press.
- Oudshoorn, Nelly, and Trevor Pinch. 2003. *How Users Matter: The Co-Construction of Users and Technology*. Cambridge, Mass.: MIT Press

Week 9 (14. Dec): Mechanical Reproduction

Reading:

- Benjamin, Walter. (1935) 1969. "The Work of Art in the Age of Mechanical Reproduction." In *Illuminations: Essays and Reflections*, edited by Hannah Arendt, translated by Harry Zohn, 217–251. New York: Schocken Books.

21. December: No Class.

Free time to work on your essay.

*****Christmas, Hanukkah, Kwanzaa, Zartosht No-Diso, New Year, etc. *****

Week 10 (11. Jan): Decolonizing Technology

Reading:

- Le Luëc, Clément, Maxime Cornet, and Antonio A. Casilli. 2023. "The Problem with Annotation. Human Labour and Outsourcing between France and Madagascar." *Big Data & Society* 10 (2): 20539517231188723.
<https://doi.org/10.1177/20539517231188723>.
- Friedner, Michele Ilana. 2022. "From Hoping to Expecting: Cochlear Implantation and Habilitation in India." *Cultural Anthropology* 37 (1): 125–49.
<https://doi.org/10.14506/ca37.1.10>.

Further Reading:

- Subramaniam, Banu, Laura A. Foster, Sandra Harding, Deboleena Roy, and Kim TallBear. 2017. "Feminism, Postcolonialism, Technoscience." In *The Handbook of Science and Technology Studies*, edited by Ulrike Felt, Rayvon Fouché, Clark A Miller, and Laurel Smith-Doerr, 407–33. Cambridge Mass.: MIT Press.
- Schramm, Katharina. 2017. "Einführung: Postkoloniale STS." In *Science and Technology Studies. Ein Überblick über Klassische Positionen und Aktuelle Perspektiven*, edited by Susanne Bauer, Thomas Lemke, and Torsten Heinemann, 471–94. Frankfurt a.M.: Suhrkamp.
- Harding, Sandra. 2009. "Postcolonial and Feminist Philosophies of Science and Technology: Convergences and Dissonances." *Postcolonial Studies* 12 (4): 401–21.
- Law, John, and Wen-yuan Lin. 2017. "Provincializing STS: Postcoloniality, Symmetry, and Method." *East Asian Science, Technology and Society* 11 (2): 211–27.

Week 11 (18. Jan): Gender and Musical Production in the Atomic Age

Reading:

- Tara Rodgers, "Toward a Feminist Historiography of Electronic Music," in *The Sound Studies Reader*, ed. Jonathan Sterne (New York: Routledge, 2012), 475–89.
- Hicks, Mar. 2021. "Sexism Is a Feature, Not a Bug." In *Your Computer Is on Fire*, edited by Thomas S. Mullaney, Benjamin Peters, Mar Hicks, and Kavita Philip, 135–158. Cambridge, MA: MIT Press.

Week 12 (25. Jan): (Artificially?) Intelligent Futures & Manifesto

Reading:

- Hau, Mark Friis, and Jakob Krause-Jensen. 2025. "Virtually Shamans: An Anthropological Perspective on AI Chatbots." *Anthropology of Consciousness* 36 (2): e70011. <https://doi.org/10.1111/anoc.70011>.
- Mohamed, Shakir, Marie-Therese Png, William Isaac, et al. "AI Decolonial Manyfesto." Manyfesto. Accessed September 9, 2026. <https://manyfesto.ai/index.html>.

Further Reading:

- Haraway, Donna Jeanne. 1991. "A Cyborg Manifesto: Science, Technology, and Socialist-Feminism in the Late Twentieth Century." In *Simians, Cyborgs, and Women: The Reinvention of Nature*. Routledge.
- Marinetti, F.T. 1909. "Declaration of Futurism." *Poesia* 5 (6). <https://bluemountain.princeton.edu/bluemtn/?a=d&d=bmtnaai190904-01.2.3>
- Su, Aaron. 2026. "AI as Pharmakon." *American Ethnologist* 53 (2): 128–31. <https://doi.org/10.1111/amet.70086>.
- LaFlamme, Marcel, and Natalie Meyers. 2026. "Enacting AI Disclosure in Scholarly Publishing." *American Ethnologist* 53 (2): 132–35. <https://doi.org/10.1111/amet.70084>.
- Menzies (hagwil hayetsk), Charles. 2026. "Appropriation, Indigenous Knowledge, and Generative AI." *American Ethnologist* 53 (2): 125–27. <https://doi.org/10.1111/amet.70081>.
- Timmer, Jaap, and Anna-Karina Hermkens. 2026. "AI as Guru or Conjurer?" *American Ethnologist* 53 (2): 136–41. <https://doi.org/10.1111/amet.70083>.
- Seaver, Nick. 2026. "On Recognizing the Handiwork of AI." *American Ethnologist* 53 (2): 142–47. <https://doi.org/10.1111/amet.70062>.

Week 13 (1. Feb): Conclusions and Feedback

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